Branch Coverage Report

Evidence generated: 30 October 2026, 23:31 IST. Baseline commit: 9ea6b597a8d51b4070b80182b056575b8c183aeb. Tests include the supplied working-tree changes.

This is an automated local verification record. It is not UAT sign-off or deployment certification. Simulated business dates in November are test inputs; execution dates are not backdated.

Measurement

Jest uses Istanbul instrumentation during the complete backend run. The denominator includes every backend/src/**/*.js source file, including models, route handlers and startup code. Frontend source is not included. Coverage measures execution, not correctness or requirement acceptance.

The source-file audit reconciles the coverage map against all 64 backend JavaScript files. Any unloaded file missing from the original Jest map receives an Istanbul zero-hit map, never assumed coverage. See evidence/coverage-scope.json for the original totals, added files and corrected totals. The Jest discovery roots were corrected after execution; a discovery-only check confirmed the identical recorded suite set for future runs.

Metric	Covered	Total	Percent
lines	1881	2031	92.61%
statements	2209	2459	89.83%
functions	466	506	92.09%
branches	1171	1454	80.53%

Per-file branch coverage

Source	Covered / total branches	Percent
backend/src/accessLog.js	6 / 12	50%
backend/src/accounts.js	18 / 23	78.26%
backend/src/app.js	20 / 23	86.95%
backend/src/archive.js	75 / 88	85.22%
backend/src/auth.js	20 / 21	95.23%
backend/src/cache.js	7 / 12	58.33%
backend/src/clock.js	0 / 0	100%
backend/src/config.js	56 / 88	63.63%
backend/src/configuration.js	189 / 255	74.11%
backend/src/entry.js	101 / 109	92.66%
backend/src/errors.js	2 / 3	66.66%
backend/src/events.js	2 / 2	100%
backend/src/exit.js	42 / 49	85.71%
backend/src/facilities.js	20 / 23	86.95%
backend/src/gates.js	25 / 35	71.42%
backend/src/hardware.js	20 / 22	90.9%
backend/src/logger.js	2 / 5	40%
backend/src/loginThrottle.js	5 / 8	62.5%
backend/src/metrics.js	6 / 12	50%
backend/src/middleware/auth.js	16 / 16	100%
backend/src/middleware/cardData.js	26 / 27	96.29%
backend/src/models/AccessLog.js	6 / 8	75%
backend/src/models/ArchiveFile.js	0 / 0	100%
backend/src/models/AuditLog.js	4 / 4	100%
backend/src/models/Counter.js	0 / 0	100%
backend/src/models/Facility.js	4 / 5	80%
backend/src/models/Gate.js	0 / 0	100%
backend/src/models/JobLock.js	0 / 0	100%
backend/src/models/LoginThrottle.js	0 / 0	100%
backend/src/models/Payment.js	11 / 12	91.66%
backend/src/models/Rate.js	2 / 2	100%

Source	Covered / total branches	Percent
backend/src/models/Receipt.js	4 / 8	50%
backend/src/models/Reservation.js	16 / 27	59.25%
backend/src/models/Session.js	15 / 19	78.94%
backend/src/models/Slot.js	1 / 2	50%
backend/src/models/User.js	2 / 2	100%
backend/src/models/Vehicle.js	0 / 0	100%
backend/src/models/Zone.js	0 / 0	100%
backend/src/notifications.js	2 / 4	50%
backend/src/occupancy.js	10 / 10	100%
backend/src/paymentGateway.js	10 / 11	90.9%
backend/src/payments.js	90 / 101	89.1%
backend/src/permissions.js	13 / 13	100%
backend/src/pricing.js	4 / 4	100%
backend/src/qr.js	12 / 12	100%
backend/src/receipts.js	11 / 12	91.66%
backend/src/reports.js	82 / 103	79.61%
backend/src/reservations.js	66 / 73	90.41%
backend/src/routes/admin.js	5 / 6	83.33%
backend/src/routes/auth.js	0 / 0	100%
backend/src/routes/events.js	6 / 6	100%
backend/src/routes/facilities.js	4 / 4	100%
backend/src/routes/gate.js	0 / 0	100%
backend/src/routes/payments.js	0 / 0	100%
backend/src/routes/reports.js	2 / 2	100%
backend/src/routes/reservations.js	0 / 0	100%
backend/src/routes/slots.js	4 / 4	100%
backend/src/routes/test.js	0 / 10	0%
backend/src/routes/users.js	5 / 6	83.33%
backend/src/server.js	0 / 8	0%
backend/src/slotSearch.js	41 / 42	97.61%
backend/src/time.js	0 / 0	100%
backend/src/transport.js	31 / 39	79.48%

Source	Covered / total branches	Percent
backend/src/validation.js	50 / 62	80.64%

Interpretation and evidence

No numeric coverage target is specified in the Part-3 assignment; none is invented here. Startup code is included even though the instrumented Jest run does not launch server.js as the production entry point. The acceptance server is a separate, uninstrumented process; its executions are not silently added to this denominator.

Open coverage/index.html for navigable source detail. coverage/coverage-final.json stores per-location execution counts; coverage/coverage-summary.json stores these aggregates.

Uncovered branch locations

Branch alternatives use zero-based Istanbul indexes. This is an improvement backlog, not a claim that an unexecuted alternative is defective.

Source	Line	Kind	Uncovered alternatives
backend/src/accessLog.js	26	if	0
backend/src/accessLog.js	30	if	0
backend/src/accessLog.js	36	if	0
backend/src/accessLog.js	40	if	0
backend/src/accessLog.js	41	if	0,1
backend/src/accounts.js	20	if	0,1
backend/src/accounts.js	35	if	0,1
backend/src/accounts.js	50	if	0
backend/src/app.js	58	if	1
backend/src/app.js	68	if	0
backend/src/app.js	71	if	1
backend/src/archive.js	55	if	1
backend/src/archive.js	75	if	0
backend/src/archive.js	82	if	0
backend/src/archive.js	101	if	1
backend/src/archive.js	159	default-arg	0
backend/src/archive.js	163	if	0
backend/src/archive.js	163	binary-expr	1
backend/src/archive.js	164	if	0
backend/src/archive.js	169	if	0
backend/src/archive.js	170	if	0

Source	Line	Kind	Uncovered alternatives
backend/src/archive.js	179	binary-expr	3
backend/src/archive.js	185	binary-expr	1
backend/src/archive.js	216	binary-expr	1
backend/src/auth.js	89	default-arg	0
backend/src/cache.js	11	cond-expr	1
backend/src/cache.js	21	if	1
backend/src/cache.js	23	if	0,1
backend/src/cache.js	24	if	0
backend/src/config.js	9	if	1
backend/src/config.js	10	cond-expr	0,1
backend/src/config.js	16	cond-expr	0
backend/src/config.js	16	binary-expr	1
backend/src/config.js	16	binary-expr	0,1
backend/src/config.js	24	binary-expr	1
backend/src/config.js	25	binary-expr	1
backend/src/config.js	26	binary-expr	1
backend/src/config.js	26	cond-expr	0,1
backend/src/config.js	26	binary-expr	0,1
backend/src/config.js	29	binary-expr	1
backend/src/config.js	30	cond-expr	0,1
backend/src/config.js	30	binary-expr	0,1
backend/src/config.js	32	cond-expr	1
backend/src/config.js	60	cond-expr	0
backend/src/config.js	67	if	0
backend/src/config.js	68	if	0
backend/src/config.js	69	if	0
backend/src/config.js	70	if	0
backend/src/config.js	71	if	0
backend/src/config.js	72	if	0
backend/src/config.js	72	binary-expr	1
backend/src/config.js	75	if	0
backend/src/config.js	75	binary-expr	1

Source	Line	Kind	Uncovered alternatives
backend/src/config.js	78	if	0
backend/src/config.js	78	binary-expr	1
backend/src/configuration.js	36	default-arg	0
backend/src/configuration.js	55	if	0
backend/src/configuration.js	64	if	0
backend/src/configuration.js	67	if	0
backend/src/configuration.js	79	if	0
backend/src/configuration.js	83	binary-expr	1
backend/src/configuration.js	84	cond-expr	1
backend/src/configuration.js	89	cond-expr	1
backend/src/configuration.js	90	if	0
backend/src/configuration.js	96	if	0
backend/src/configuration.js	99	if	0
backend/src/configuration.js	103	binary-expr	1
backend/src/configuration.js	106	cond-expr	0,1
backend/src/configuration.js	107	if	0
backend/src/configuration.js	108	if	0
backend/src/configuration.js	112	default-arg	0
backend/src/configuration.js	114	if	0
backend/src/configuration.js	115	if	0
backend/src/configuration.js	116	if	0
backend/src/configuration.js	121	if	0
backend/src/configuration.js	124	if	0
backend/src/configuration.js	125	binary-expr	1
backend/src/configuration.js	127	cond-expr	1
backend/src/configuration.js	132	if	0
backend/src/configuration.js	133	binary-expr	1
backend/src/configuration.js	137	binary-expr	1
backend/src/configuration.js	139	cond-expr	1
backend/src/configuration.js	141	binary-expr	1
backend/src/configuration.js	154	if	0
backend/src/configuration.js	194	default-arg	0

Source	Line	Kind	Uncovered alternatives
backend/src/configuration.js	219	if	0,1
backend/src/configuration.js	224	default-arg	0
backend/src/configuration.js	227	if	0
backend/src/configuration.js	233	binary-expr	1
backend/src/configuration.js	239	default-arg	0
backend/src/configuration.js	241	if	0
backend/src/configuration.js	243	if	0
backend/src/configuration.js	250	default-arg	0
backend/src/configuration.js	253	if	0
backend/src/configuration.js	267	cond-expr	1
backend/src/configuration.js	267	cond-expr	1
backend/src/configuration.js	271	default-arg	0
backend/src/configuration.js	275	if	0
backend/src/configuration.js	275	binary-expr	1
backend/src/configuration.js	280	if	0
backend/src/configuration.js	297	if	0,1
backend/src/configuration.js	300	if	0
backend/src/configuration.js	308	default-arg	0
backend/src/configuration.js	311	if	0,1
backend/src/configuration.js	319	default-arg	0
backend/src/configuration.js	321	if	0,1
backend/src/configuration.js	328	default-arg	0
backend/src/configuration.js	329	cond-expr	1
backend/src/configuration.js	334	default-arg	0
backend/src/configuration.js	336	if	0
backend/src/configuration.js	337	if	0
backend/src/configuration.js	338	if	0
backend/src/configuration.js	346	cond-expr	1
backend/src/configuration.js	347	if	0
backend/src/configuration.js	349	binary-expr	1
backend/src/configuration.js	352	binary-expr	1
backend/src/entry.js	36	default-arg	0

Source	Line	Kind	Uncovered alternatives
backend/src/entry.js	41	if	0
backend/src/entry.js	45	if	0
backend/src/entry.js	75	if	0
backend/src/entry.js	189	if	0
backend/src/entry.js	214	if	0,1
backend/src/entry.js	257	default-arg	0
backend/src/errors.js	16	default-arg	0
backend/src/exit.js	21	default-arg	0
backend/src/exit.js	23	if	0
backend/src/exit.js	26	if	0
backend/src/exit.js	37	cond-expr	1
backend/src/exit.js	41	cond-expr	1
backend/src/exit.js	80	cond-expr	0
backend/src/exit.js	93	if	1
backend/src/facilities.js	17	default-arg	0
backend/src/facilities.js	36	binary-expr	1
backend/src/facilities.js	40	binary-expr	1
backend/src/gates.js	21	default-arg	0
backend/src/gates.js	24	if	0
backend/src/gates.js	25	if	0
backend/src/gates.js	26	binary-expr	1
backend/src/gates.js	30	if	0
backend/src/gates.js	32	cond-expr	1
backend/src/gates.js	33	cond-expr	1
backend/src/gates.js	47	default-arg	0
backend/src/gates.js	48	cond-expr	1
backend/src/gates.js	54	default-arg	0
backend/src/hardware.js	37	if	0
backend/src/hardware.js	48	if	0
backend/src/logger.js	7	if	1
backend/src/logger.js	9	cond-expr	0,1
backend/src/loginThrottle.js	23	if	0

Source	Line	Kind	Uncovered alternatives
backend/src/loginThrottle.js	37	if	0
backend/src/loginThrottle.js	49	if	1
backend/src/metrics.js	15	if	0
backend/src/metrics.js	16	if	0
backend/src/metrics.js	27	cond-expr	1
backend/src/metrics.js	34	cond-expr	1
backend/src/metrics.js	36	cond-expr	1
backend/src/metrics.js	37	cond-expr	0
backend/src/middleware/cardData.js	26	if	0
backend/src/models/AccessLog.js	19	binary-expr	1
backend/src/models/AccessLog.js	19	binary-expr	1
backend/src/models/Facility.js	15	default-arg	0
backend/src/models/Payment.js	40	binary-expr	1
backend/src/models/Receipt.js	28	binary-expr	1
backend/src/models/Receipt.js	29	binary-expr	1
backend/src/models/Receipt.js	29	binary-expr	1
backend/src/models/Receipt.js	33	binary-expr	1
backend/src/models/Reservation.js	33	default-arg	0
backend/src/models/Reservation.js	37	binary-expr	1
backend/src/models/Reservation.js	38	binary-expr	1
backend/src/models/Reservation.js	39	binary-expr	1
backend/src/models/Reservation.js	39	binary-expr	1
backend/src/models/Reservation.js	40	binary-expr	1
backend/src/models/Reservation.js	40	binary-expr	1
backend/src/models/Reservation.js	41	binary-expr	1
backend/src/models/Reservation.js	41	binary-expr	1
backend/src/models/Reservation.js	41	binary-expr	1
backend/src/models/Reservation.js	42	binary-expr	1
backend/src/models/Session.js	38	default-arg	0
backend/src/models/Session.js	42	binary-expr	1
backend/src/models/Session.js	43	binary-expr	1
backend/src/models/Session.js	44	binary-expr	1

Source	Line	Kind	Uncovered alternatives
backend/src/models/Slot.js	25	binary-expr	1
backend/src/notifications.js	10	default-arg	0
backend/src/notifications.js	12	if	0
backend/src/paymentGateway.js	24	cond-expr	0
backend/src/payments.js	26	default-arg	0
backend/src/payments.js	71	cond-expr	0
backend/src/payments.js	78	cond-expr	1
backend/src/payments.js	107	if	0
backend/src/payments.js	115	if	0
backend/src/payments.js	126	if	0
backend/src/payments.js	150	if	1
backend/src/payments.js	158	default-arg	0
backend/src/payments.js	159	if	0
backend/src/payments.js	194	default-arg	0
backend/src/payments.js	197	binary-expr	1
backend/src/receipts.js	20	if	0
backend/src/reports.js	38	default-arg	0
backend/src/reports.js	47	if	0
backend/src/reports.js	77	if	0
backend/src/reports.js	110	binary-expr	1
backend/src/reports.js	119	binary-expr	1
backend/src/reports.js	128	cond-expr	1
backend/src/reports.js	129	cond-expr	1
backend/src/reports.js	137	cond-expr	1
backend/src/reports.js	138	cond-expr	1
backend/src/reports.js	157	if	1
backend/src/reports.js	162	cond-expr	0
backend/src/reports.js	167	if	1
backend/src/reports.js	170	binary-expr	1
backend/src/reports.js	176	if	0
backend/src/reports.js	179	if	0
backend/src/reports.js	180	if	0

Source	Line	Kind	Uncovered alternatives
backend/src/reports.js	181	if	1
backend/src/reports.js	185	cond-expr	1
backend/src/reports.js	203	binary-expr	1
backend/src/reports.js	206	if	0
backend/src/reports.js	207	cond-expr	0
backend/src/reservations.js	28	default-arg	0
backend/src/reservations.js	32	if	0
backend/src/reservations.js	33	if	0
backend/src/reservations.js	50	if	0
backend/src/reservations.js	89	binary-expr	1
backend/src/reservations.js	107	binary-expr	1
backend/src/reservations.js	108	if	0
backend/src/routes/admin.js	77	if	0
backend/src/routes/test.js	27	binary-expr	0,1
backend/src/routes/test.js	28	binary-expr	0,1
backend/src/routes/test.js	30	binary-expr	0,1
backend/src/routes/test.js	32	binary-expr	0,1
backend/src/routes/test.js	48	binary-expr	0,1
backend/src/routes/users.js	44	if	0
backend/src/server.js	20	if	0,1
backend/src/server.js	21	if	0,1
backend/src/server.js	21	binary-expr	0,1
backend/src/server.js	25	if	0,1
backend/src/slotSearch.js	39	if	0
backend/src/transport.js	43	binary-expr	1
backend/src/transport.js	44	if	0
backend/src/transport.js	55	binary-expr	1
backend/src/transport.js	78	binary-expr	1
backend/src/transport.js	79	if	0
backend/src/transport.js	83	cond-expr	0
backend/src/transport.js	94	if	0
backend/src/transport.js	102	if	1

Source	Line	Kind	Uncovered alternatives
backend/src/validation.js	23	cond-expr	1
backend/src/validation.js	42	if	0
backend/src/validation.js	51	default-arg	0
backend/src/validation.js	59	if	0
backend/src/validation.js	78	default-arg	0
backend/src/validation.js	83	if	0
backend/src/validation.js	83	binary-expr	1
backend/src/validation.js	89	binary-expr	1
backend/src/validation.js	93	default-arg	0
backend/src/validation.js	102	default-arg	0
backend/src/validation.js	112	default-arg	0
backend/src/validation.js	120	default-arg	0